

How We Tested

DVD WRITERS

Our test-bed for DVD burners comprised an Intel Pentium 4 3.0 GHz on an MSI 875P NEO motherboard, supported by 512 MB of DDR400 SDRAM and a 200 GB Seagate Barracuda SATA hard disk. Microsoft Windows XP with SP1, along with the latest drivers, was the operating system used.

Features

Features such as Mt Rainer support, data buffer, silent operation, etc., were rated on a scale of five.

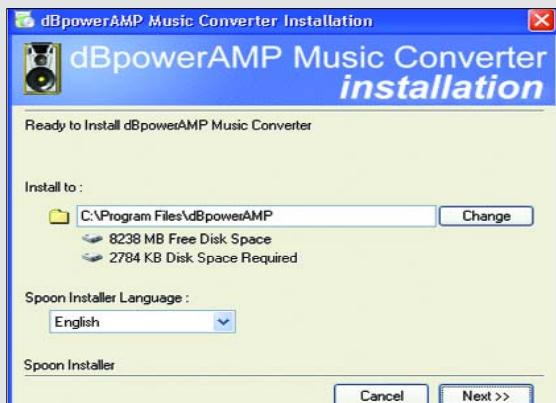
Package Content

Here we checked and allocated points to accessories that added value, such as data cable and blank DVD media, provided along with a drive. Similarly, a drive with DVD-authoring and playback software was awarded more points.

Performance

We ran a battery of tests—a mix of synthetic and real-world—to gauge the performance of the drive. Nero Burning ROM 6.3.1.20 was used as the authoring software to burn data on CDs and DVDs.

Nero Speed: Nero Speed was used to measure transfer rate and data throughput, in addition to access time and CPU utilisation of the drives in question. This test offers insight on a drive's performance.



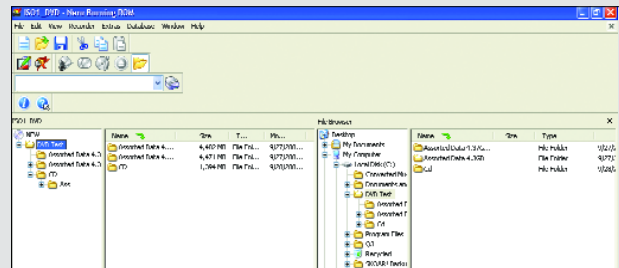
We used dbPowerAMP to test CD Ripping

SiSoft Sandra 2004: SiSoft Sandra is a synthetic benchmark that measures the sequential and random read speed of a drive. It does this by constantly reading data at the maximum speed supported. It also measures the access time.

CD Ripping Test Using dbPowerAMP: In this test, we encoded an audio CD of 13 tracks and noted down the time taken by each drive.

DVD Video Extraction Using DVD Decrypter: Using DVD Decrypter 3.5.1.0, we encoded a DVD movie and noted down the time taken by each drive to complete the task.

Burning Test For A CD/CD-RW: In this test we burned 700 MB of assorted data and a 700 MB file on a CD/CD-RW. The time taken by each drive was then noted. We also noted the time needed to erase a re-writable disc. We used 52X Moser Baer CD-R and Mitsubishi 24X US (Ultra Speed) CD-RW as optical media.



Nero Burning ROM was used to burn DVDs

Burning Test For A DVD: A 4.3 GB file and assorted data were written on a Memorex 8X DVD-R, a Verbatim 8X DVD+R and a Sony 4X DVD+RW media. We ran additional erasing tests for DVD+RW media and noted the time taken. For the dual-layer drives we used a Verbatim 2.4X and noted down the time taken to burn 8.5 GB data on it.

EXTERNAL HARD DRIVES

Our test configuration, consisted of an Intel Pentium 4 2.6 GHz processor on an i845 chipset-based, ABit BG7E motherboard. The rig had 256 MB of DDR SDRAM and a Seagate 40 GB hard disk. The latest drivers for the motherboard and USB ports were loaded. To maintain a standard environment, we installed only the required benchmarking utilities and the needed test files.

Features

We looked at various inbuilt characteristics and qualities of each drive. Considering that every product was marketed as a portable device, we laid emphasis on a drive's weight.

Portability was awarded less importance for a 3.5-inch drive, since their usage will primarily be as a stationary backup device.

Build-quality was also assessed, with particular importance given to the sturdiness of the product. Bonus points were awarded for miscellaneous features such as extra USB/FireWire port, configurable buttons, and also for useful software included.

Performance

Both synthetic and real-world tests were conducted to gauge performance. SiSoft Sandra Professional 2004 was used to assess theoretical read and write speeds, as well as the average access time. We also tested with HDTach 2.7 and noted the average CPU utilisation of each drive.

For real-world performance evaluation, we transferred a set of assorted files of varying sizes, totalling 1 GB, back and forth, from the system hard disk to the external hard disk, and noted the time taken during each transfer. Similarly, to test the sequential speeds, we transferred a large image of 1 GB between the system and the external drive.

Price Index

This was calculated taking into account the ideal price for which an external hard disk is available.

How We Awarded

Scores granted to Features, Build Quality, Performance and Price were given weightages relevant to specific categories. An overall score out of 100 was then calculated. The product that scored the highest here was adjudged the winner of the Digit Best Buy Gold award for its category. The runner-up was awarded the Digit Best Buy Silver award.